

FIDELITYLOOP: CLOSED-LOOP VALIDATION OF ELASTIC LLM-SERVING POLICIES

Anonymous Authors¹

ABSTRACT

Elastic LLM-serving policies depend on when capacity becomes available, not just on request service times. We present and implement FidelityLoop, a closed-loop validation framework connecting replay predictions, physical lifecycle execution, and deadline-aware resource accounting. In a preregistered study of 36 Qwen2.5-7B-Instruct runs on two A100 GPUs, the calibrated replay improves matched-request timing while worsening occupied-resource prediction; trajectory analysis explains why aggregate error is insufficient for closed-loop evaluation. We then train PPO scheduling policies in replay under two startup-time assumptions and freeze five checkpoints for a 32-run deployment campaign (E2). Three checkpoints meet every deadline in all six runs; two fail all six. Both training environments produce passing and failing checkpoints. Replay with calibrated service times and the historical startup proxy (C) predicts feasibility correctly on 32/32 runs, versus 20/32 after changing startup to 30 seconds (C30). Their cost MAEs are 0.0804 and 0.4657 scenario USD. A follow-up on six new checkpoints instead finds all six feasible, including three rejected by C. These cases show why both policies and their replay models need deployment validation: aggregate resource error and past feasibility agreement can each obscure wrong decisions. FidelityLoop exposes these failures under a shared policy contract and releases paired traces for reproducible analysis within the studied workload, hardware, and synthetic-API setting.

1 INTRODUCTION

Elastic LLM serving releases idle instances to save resources, but capacity returns only after startup. Queues, running work, and lifecycle transitions then shape subsequent control decisions. A simulator used to evaluate such policies must capture when capacity is available, not just how long individual requests take or how many GPU seconds a run consumes.

FidelityLoop studies these requirements as a closed loop from prediction to policy decision to physical lifecycle execution. In 36 preregistered runs, the calibrated replay improves matched-request timing yet worsens occupied-resource prediction under the same hysteresis rule (H). A startup-endpoint contrast locates one feedback path at the controller’s cooldown boundary, while trajectory analysis shows that equal total occupation error can hide different control inputs. We then train PPO policies in replay under two startup-time assumptions, freeze five checkpoints, and evaluate them against an always-on two-GPU reference. This 32-cell campaign uses the same registered Qwen/A100

serving stack under a separately re-provisioned runtime lock to test whether predicted feasibility survives physical execution. A further six-checkpoint screening study tests whether the better-performing replay model remains useful for new candidates; its false rejections expose a limit that aggregate agreement alone misses.

The paper makes three contributions:

1. **A closed-loop evaluation framework.** FidelityLoop separates the policy adapter from lifecycle execution and aligns causal observations, request identities, lifecycle generations, resource trajectories, and deadline-aware cost ledgers. The same contract runs fixed rules and learned checkpoints, with failed requests retained in the denominator.
2. **A lifecycle-fidelity diagnosis.** Held-out physical runs distinguish local-service, aggregate-resource, and trajectory fidelity. Event-level comparisons explain a startup-related divergence and show why aggregate resource error is insufficient for a closed-loop evaluator.
3. **A reproducible policy evaluation.** A guard for pending offline deadlines quantifies the resource–latency trade-off, and the continuous E2 campaign tests five frozen PPO checkpoints under two arrival regimes. A follow-up screening study exposes false rejections of feasible new

¹Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. Correspondence to: Anonymous Author <anon.email@domain.com>.

checkpoints. These cases test both policies and their evaluator: lower scenario cost is not a saving when deadlines are missed. We release request-aligned prediction and execution traces with CPU analysis tools.

The release provides CPU interface checks, a fixed-event cost verifier, and a paired trace corpus; the supplement documents their reproduction scopes.

2 BACKGROUND AND MOTIVATION

We consider mixed online and offline LLM workloads served by a limited GPU pool, where elastic scheduling aims to reduce resource use while meeting request deadlines. Evaluating every candidate policy on physical hardware is costly, motivating a framework for replay-based policy development and training followed by physical validation. FidelityLoop connects these stages through a shared policy interface and consistent lifecycle and deadline accounting, so that savings predicted in replay can be checked against deployment outcomes.

Our exploratory deployment revealed why this connection matters: the calibrated replay reduced prediction error for matched local requests while increasing error in occupied GPU time under dynamic scheduling. These preliminary observations motivate the preregistered study; they are not additional held-out repeats.

The lifecycle diagnosis separated the old 69.219 s startup proxy from the earlier window launch-to-ready (25.19 s) and setup (46.80 s) contexts. When the proxy crossed the 60 s cooldown boundary it changed a target decision, motivating the locked sequence of calibration, lifecycle checks, and held-out feasibility tests.

Figure 1 summarizes the lifecycle boundary. It is a motivating diagnostic, not a post-hoc selection rule: the held-out windows were selected once from the registered forward pool and were not replaced after observing performance.

3 RELATED WORK

Vidur and Charon use performance simulation to guide LLM configuration choices; Clockwork studies predictable request timing (Agrawal et al., 2024; Yang et al., 2026; Gujarati et al., 2020). FidelityLoop complements these efforts with a shared policy interface for replay and physical execution, aligning request identities, lifecycle trajectories, and deadline-aware resource ledgers. This lets us test whether better service-time fit also improves resource prediction and policy feasibility.

Physical serving systems change how capacity becomes available: cold-start characterization and systems such as ServerlessLLM expose this lifecycle cost (Kabakibo et al.,

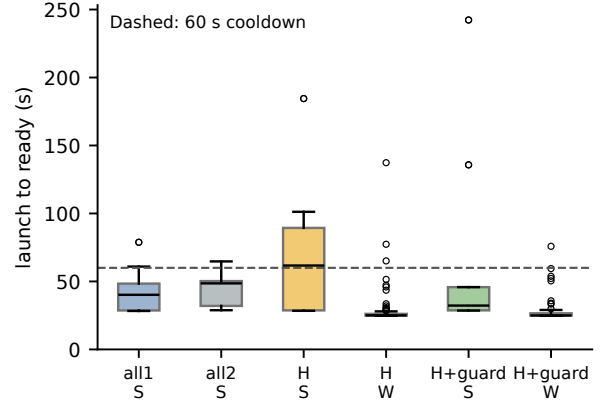


Figure 1. Observed startup times in the held-out runs. Labels use S for setup and W for window scale-out. The dashed line marks the 60-second controller cooldown; boxes summarize observed launch-to-ready times. Release and cleanup are separate ledger events. Collapsing these contexts into one endpoint can alter closed-loop actions.

2026; Fu et al., 2024). We hold the engine and loading path fixed, changing only simulated timing in the endpoint comparison.

Provisioning and scheduling work connects workload estimates to SLOs (Crankshaw et al., 2020; Zhang et al., 2019; 2023). FidelityLoop provides a shared evaluation contract for such policies, connecting replay predictions to physical execution and deadline-aware resource accounting. The guard serves as a rule-based case study; frozen PPO checkpoints extend the evaluation to learned policies.

Policy design provides a complementary layer. Llumix dynamically reschedules LLM requests (Sun et al., 2024); AAPA predicts scaling needs for serverless workloads (Zhang et al., 2025b); KIS-S learns GPU-inference autoscaling with PPO (Zhang et al., 2025a). We use PPO as a frozen policy family rather than proposing a new RL algorithm; PPO is the standard clipped policy-gradient method (Schulman et al., 2017). Our question is whether a learned checkpoint selected in replay remains feasible after the lifecycle backend and deadline ledger are applied.

4 PROBLEM AND METRICS

The system has two accelerators, online arrivals with a 60 s deadline, and 120 offline jobs per 2,100 s observation window with a 300 s deadline. Online requests run locally or use a delay-only synthetic API sink; offline jobs run locally. Arrivals span 1,800 s; execution and accounting continue for another 300 s to cover the remaining deadline

horizon. The primary window cost, in scenario USD, is

$$C = .001G + .002S + .001D + 10^{-6}I + 3 \times 10^{-6}B + .01M, \quad (1)$$

where G is occupied GPU seconds, S, D are starts and shutdowns, I, B are synthetic API input and output budgets, and M counts offline deadline misses, including unfinished requests whose deadlines have passed. Setup and cleanup are reported separately and are not folded into the primary model gate.

Completed means a recorded completion by the observation cutoff; it may be *on time* (finish time $\leq$ deadline) or *late* (finish time $>$ deadline). An unfinished request has no completed terminal by the cutoff. Unfinished or censored work is retained in the arrival denominator. The capacity gate G_F requires all2 (two GPUs throughout) to complete every online and offline request on time. P1 requires 120/120 offline requests on time, at least 99% online requests on time, and no unfinished request; P1+ additionally requires every online request on time. P2 is a ratio-of-means saving against all1 (one GPU throughout) computed within each window and then averaged equally across the three windows.

5 WORKLOAD CONSTRUCTION

Online arrivals in the main study come from the successful-request subset of BurstGPT, a public trace of production LLM requests (Wang et al., 2024). We read all 1,404,294 source rows and retain source hashes and row identities. The trace has no session identifier, so the split separates arrival times rather than sessions. Locally routed requests execute actual GPU inference with fixed-corpus prompts and registered token budgets. The experiment reproduces arrival-time patterns, not the original users’ prompts or semantic tasks.

The three held-out windows are selected once from the forward pool after the old windows plus a 30-minute protection band. A 60 s morphology filter defines the steady, recovery, and burst-offline classes (even, declining, and bursty online arrivals); a fixed SHA ordering takes the first eligible candidate in steady–recovery–burst order, subject to the registered separation constraint. Online rows are thinned with the inherited C2 integer threshold, while the 120 offline jobs are constructed at fixed 15 s spacing and 256-input/256-output token budgets. The two salts and all thresholds are fixed in `s4_generation_lock.json`; generated inputs are locked before replay and never replaced to improve a result.

Table 1 reports the arrival shape at the registered 60 s scale. The 1 s diagnostics are retained in the manifest but do not define the class. The values describe the selected derived windows, not the full source population.

The derived windows preserve arrival timing and fixed work

Table 1. Selected window arrival shape at 60 s bins. Each cell is raw/derived; CV and peak-to-mean use the full 1,800 s window, including empty bins.

Window	arrivals	CV	peak/mean	lag-1
steady	5461 / 1444	0.378 / 0.412	1.566 / 1.641	0.772 / 0.797
recovery	1141 / 334	1.444 / 1.452	3.550 / 3.862	0.906 / 0.892
burst-offline	663 / 177	1.951 / 1.935	6.652 / 6.780	0.810 / 0.754

accounting, but they are late-source slices with no session fields. They therefore represent a controlled workload domain for lifecycle decisions, not a claim about all production traffic or user intent.

E2 uses two separately generated and locked windows from the same construction pipeline, with no row overlap with the three held-out windows; the campaigns remain distinct and are never pooled. The recovery window contains 358 derived online arrivals and 120 fixed offline arrivals (478 total); the steady window contains 944 online and 120 offline arrivals (1,064 total). Offline arrivals are spaced every 15 s, with 256 input and output tokens; the online rows retain their locked arrival timestamps. The window identities and input hashes are bound in the E2 lock.

A separate external-validity study uses Azure conversation arrival traces associated with Splitwise (Patel et al., 2024); its workload construction and load-boundary results are reported in the supplement.

6 EXPERIMENTAL SETUP

The preregistered held-out study used two NVIDIA A100 80GB PCIe GPUs at a university computing facility, with one independent tensor-parallelism-1 instance per GPU and Qwen2.5-7B-Instruct. The serving stack was vLLM 0.9.0 (Kwon et al., 2023) with PyTorch 2.7.0+cu126, Transformers 4.51.1, tokenizers 0.21.1, and NumPy 1.26.4. Instances used FP16, maximum model length 4096, maximum batched tokens 16,384, maximum sequences 1,024, four local request slots, chunked prefill enabled, prefix caching disabled, eager execution enabled, and GPU memory utilization 0.8. Readiness required the per-process health probe and the dual-ready barrier; cleanup required verified process release on both devices. These values are bound by each run’s environment snapshot and GPU identity record.

The held-out study and E2 both use two A100 GPUs. Before held-out execution, the deployment-adapted model (E) had its lifecycle parameters calibrated and frozen. The historical V–E comparison spans different hardware environments (A800 and A100), so it does not isolate the effect of calibration.

The E2 campaign instantiated the same registered Qwen2.5-7B-Instruct serving stack on two A100 GPUs through the framework contract. It contains 32 locked cells: one all2 capacity anchor and five frozen PPO arms per window, with three repeats per comparison arm across recovery and steady windows. The campaign was run under a new runtime lock, with each checkpoint and tensor identity bound into the result package. It is a policy-execution test, not a model-selection search or a claim of universal PPO superiority. An earlier 38-run campaign, termed Bridge, evaluated rule-based, guarded, and PPO policies under a separate runtime lock; its results are retained as supplementary evidence and are not pooled with E2. The follow-up screening study (E3) uses six previously undeployed checkpoints on the same two windows. Each window runs on a separate dual-A100 allocation with the registered serving stack: one all2 anchor and two repeats per arm, 26 cells in total. E3 is analyzed separately from E2 and Bridge.

7 METHOD

Predictive replay and physical execution share a policy contract. Adapters read observable queues and device states and return capacity targets; the guard may also select dispatch priority. The backend handles readiness, draining, and verified release, while request identities align predictions, decisions, lifecycle states, resources, and costs. The resulting data flow is *workload rows* $\rightarrow$ *causal observation* $\rightarrow$ *policy adapter* $\rightarrow$ *lifecycle backend* $\rightarrow$ *identity ledger*. A separate CPU verifier recomputes feasibility and cost from raw records. By default, ready local slots serve online requests before offline jobs. Online requests still waiting 10 s after arrival can enter the synthetic API sink, which admits up to eight concurrent requests and completes each on a 5 s timer. The guard can prioritize at-risk offline jobs; PPO changes capacity while retaining the default dispatch and API rules.

In the frozen physical stack, the queue-target, predictive, and queueing-capacity baselines each use 23–38 policy-specific code lines and share a 73-line adapter and support layer. The supplement separates policy logic, integration code, and reused dependencies for these baselines, the guard, and PPO.

The CPU framework exposes interfaces for replacing performance estimators and policies. On small synthetic workloads, four historical policy configurations preserve requests, events, and ledgers exactly, and two device/slot configurations pass conservation checks. Complementing these interface tests, causal event replay reconstructs observations and policy state and reproduces all 29,580 logged decisions from 16 earlier same-runtime runs (supplement). This validates decision reconstruction under recorded events; prediction fidelity is evaluated separately below.

Table 2. Frozen replay parameters in seconds, rounded to three decimals. All models share a 69.363 s setup proxy outside the primary window ledger.

Model	Service	Startup	Shutdown
O	equal-share	30.000	7.059
C	measured lookup	69.219	7.059
C30	same lookup	30.000	7.059
E	same lookup	33.497	2.224
V	same lookup	25.196	2.325

7.1 Calibrated models

The replay models share an executor but differ in service and lifecycle timing (Table 2). O uses an equal-share service model and an assumed 30 s startup. C uses a measured service lookup and a historical 69.219 s startup proxy; C30 restores the 30 s assumption. E combines the lookup with deployment-adapted lifecycle parameters; V retains a bounded refit from the earlier environment. O–C30 isolates the service model, whereas C–C30 isolates startup; O–C changes both.

O assumes 1024 input and 128 output tokens/s, shared equally among concurrent requests. The calibrated lookup uses median measured wall times per input/output/concurrency cell, interpolating in input length and concurrency (one to four requests per instance). Its values are shared by C, C30, E, and V. The supplement’s Replay model specification section details measurement sources, interpolation, and lifecycle fitting. Model bytes, tokenizer, workload identities, and the candidate grid are locked before held-out execution.

For each model prediction and physical repeat of a window, the dimensionless lifecycle loss is

$$L = \frac{0.5|\Delta G_{\text{occupied}}| + 0.25|\Delta G_{\text{active}}|}{2T} + \frac{0.25|\Delta Q|}{TN}, \quad (2)$$

Here G_{occupied} sums per-GPU time from launch issuance to verified release, while G_{active} sums time in the active state, including idle time; both are measured in GPU seconds. The difference Δ is prediction minus observation, $T = 2100$ s for formal windows, and N counts *all* online and offline arrivals. Q is the time integral of the released-but-undispatched queue, in request seconds. Unfinished requests remain in N and contribute queue time while queued up to cutoff. Two devices give the $2T$ normalization. We first average losses over repeats within each window, then weight windows equally; we do not pool requests or restrict N to matched-local support. L measures prediction error, separately from scenario cost C ; the shorter development horizon is stated with that experiment.

7.2 PPO policy contract

The learned policy is trained only in the CPU replay environment, with C or C30 as the training environment; the physical serving loop is used only after a checkpoint is frozen. At each one-second decision boundary, its 30-dimensional causal observation contains normalized time and target state, queue counts and ages/slack, device lifecycle states and ages, running-job summaries, and the API queue. The capacity action proposes a target of 0, 1, or 2 GPUs; the registered 60 s cooldown filters that proposal before execution, while the offline action head is fixed to its sole legal value. The reward is the negative incremental scenario-ledger cost, so training trades occupied resources, lifecycle events, API use, and deadline misses under the declared contract.

Training uses the three development arrival regimes and paired seeds from the locked CPU PPO contract. The five E2 checkpoints are selected mechanically with C30 replay, guard disabled, P1+ required in every validation window, and sorting by equal-window mean cost, checkpoint index, then seed. We label the frozen checkpoints C-1/C-2 and C30-1/C30-2/C30-3 by training environment, numbering them by seed and then episode, independently of physical outcomes. Training settings, data partitions, and checkpoint coordinates are detailed in the supplement’s Frozen PPO training and selection contract section.

7.3 Controller and guard

H uses up-threshold 8, down-threshold 0, and a 60 s cooldown. At each one-second tick, H reads queue and running counts plus its target/cooldown state, not the predictor’s startup parameter. The guard’s lifecycle-delay estimates use separately frozen safety parameters, distinct from the C/E simulator endpoints. The guard observes only released queues, device states, and running-job ages; future arrivals and true completion or readiness times are unavailable. Let P_t be the pending offline request IDs and b_t H’s current target after its decision. The projection $F_j(k, r)$ estimates request j ’s finish with k devices and reservation setting r . It chooses devices by estimated lifecycle delay (device ID breaks ties), creates four slots per device, and includes running residuals. Service durations use the frozen concurrency-four lookup. An overdue running estimate falls back to one positive full-service duration; starting and stopping devices retain lifecycle delay rather than becoming instantaneous capacity.

Offline jobs are ordered by deadline then request ID. Without reservation, visible online backlog precedes them in the list schedule. With reservation and online backlog, offline work is projected serially from the earliest running offline slot, or the earliest slot if none is running; other slots remain online-first. With no online backlog, offline jobs use all projected slots. These are conservative estimates, not proved

completion-time upper bounds.

For the full guard ($r = 1$), margin $m = 5$ s and previous risk-ID set A_{t-1} , the update is

$$\begin{aligned} R_t &= \{j \in P_t : F_j(\max(1, b_t), 1) + m + 1 \geq d_j\}, \\ A_t &= (A_{t-1} \cap P_t) \cup R_t, \\ K_t &= \{k \in \{1, 2\} : \forall j \in P_t, F_j(k, 1) + m \leq d_j\}, \\ u_t &= \begin{cases} b_t, & A_t = \emptyset, \\ \max(b_t, \min K_t), & |A_t| > 0, |K_t| > 0, \\ 2, & |A_t| > 0, K_t = \emptyset. \end{cases} \end{aligned} \quad (3)$$

Here d_j is the deadline. Triggering includes a one-second lookahead and uses $\geq$; candidate feasibility uses $\leq$ without that extra second. Empty K_t with active risk sets an `infeasible` flag. A latched ID is removed only when it leaves the pending queue, not when a later estimate becomes safe. The target never falls below *this tick’s* H target. A guard raise can bypass H’s cooldown and writes u_t and the current time back to H’s target and last-change state; it does not guarantee global deadline feasibility.

While A_t is nonempty, dispatch reserves at most one concurrent offline slot when both queues have work. A running offline job leaves other free slots online-first. Otherwise, with head-of-queue slacks $s_x = d_x - t - \hat{p}_x$, online wins if $s_{\text{on}} \leq m$ and $s_{\text{on}} \leq s_{\text{off}}$; offline wins otherwise. With no online backlog, offline work may use all free slots. No running request is preempted. Target changes therefore preserve in-flight work, and later scale-out pays lifecycle delay before providing capacity.

The development search considered the registered hysteresis family and its guard variants, including 26 candidate controller configurations. Selection used the development contract and feasibility gates before the held-out inputs were read. The selected guarded configuration was identical to the preregistered H+guard arm, so the duplicate P_{selected} arm was merged into the four-arm formal matrix to avoid pseudo-replication.

7.4 Evidence protocol

The held-out study evaluates all1, all2, H, and H+guard on steady, recovery, and burst-offline windows, with three repeats per arm (36 formal runs). The GPU package is verified again on CPU using the original acceptance code and the lock-bound source snapshots. Two operator-aborted attempts are retained as history and excluded from formal statistics. All metrics are reported at run and window levels; requests are not treated as independent workloads. For Bridge, the CPU receiver independently recomputes terminal counts, lifecycle occupation, and window scenario cost from raw event streams; the sealed cell summaries are used only for consistency checks. A technically valid negative

result remains in the matrix.

The gate order is fixed before looking at held-out outcomes. G_F first asks whether the two-instance reference can finish all registered work on time. M1 then compares occupied-resource error against the original model, M1b compares the lifecycle objective after deployment adaptation, and M2 checks local service fidelity on matched request IDs. M3 is a diagnostic target-agreement statistic rather than a replacement for the service and occupancy gates. Only after these model gates are interpreted do P1/P1+ test online and offline feasibility and P2 test cost. This ordering prevents an attractive cost result from rescuing a model that cannot represent the service contract, while still allowing H's failure to remain a useful ablation instead of stopping the entire matrix.

8 RESULTS

We first establish the prediction-level mismatch that motivates the framework. We then run the deployment campaign—the central test of whether replay feasibility survives physical lifecycle execution—and finish with the guard control and its resource–latency boundary.

8.1 Cross-level prediction fidelity

On fixed H, E's occupied MAE is 67.94 GPU seconds versus 543.51 for O (87.50% lower). C30's improvement over C implicates startup. V reduces equal-window L by 34.24% relative to E, but E uses deployment-adapted endpoints while V retains the earlier environment snapshot. M1b is therefore not a clean same-environment causal comparison.

For local service fidelity, the fixed all1/all2 tests give a maximum window MAE of 0.064 s for C, C30, E, and V, while O reaches 3.016 s. On dynamic arms, the minimum common support across model/route projections is 122 requests and the maximum common-support MAE is 0.129 s for C, 0.112 s for C30, 0.110 s for E, and 0.112 s for V; O reaches 4.510 s. Common support is the intersection of request IDs completed locally in the compared predictions and physical runs; coverage and routing discrepancies are reported in the artifact.

Table 3 reports window-wise occupied-resource errors for fixed-H replay, distinct from the route-conditioned matched-ID checks.

The C-versus-O comparison reproduces the exploratory divergence: maximum matched-ID service MAE falls from 4.510 to 0.129 s (about 35-fold), while equal-window occupied-resource MAE rises from 543.51 to 891.98 GPU seconds (64.1%). The aggregate deterioration does not occur in every window. O–C changes both service and startup; the C–C30 contrast isolates startup, lowering occupied MAE

Table 3. Occupied-resource MAE (GPU seconds) by window and model. L is the equal-window lifecycle loss defined above.

Model	steady	recovery	burst	Occupied MAE	L
O	1151.97	244.08	234.48	543.51	0.09403
C	964.14	864.20	847.60	891.98	0.12162
C30	67.18	78.30	83.08	76.19	0.01353
E	66.34	66.95	70.55	67.94	0.01353
V	43.51	65.74	69.34	59.53	0.00890

Table 4. Steady-window fidelity: means over the same three physical repeats. Resource errors are GPU seconds; target disagreement is a percentage of decision ticks.

Model	Total error	Trajectory error	Target diff. (%)
C30	67.18	114.32	1.03
E	66.34	1390.50	66.73

to 76.19 GPU seconds.

Retrospective analysis of the frozen trajectories explains this endpoint effect and its limits (supplement). In all three windows, C/C30 targets first diverge 60 s after a common instance start: C remains unready with higher queue/running pressure; C30 is ready. This simulator contrast is not a physical intervention on startup or cooldown. Six burst/recovery physical repeats first disagree in target with predictions at 3 s, before any window restart. Conversely, steady repeats 1 and 3 match C30 targets throughout despite state, routing, and occupation errors.

Aggregate agreement can mask temporal misalignment. In steady, E and C30 have nearly equal total occupation MAE, but E's mean time-integrated absolute error in occupied GPU count is 1390.504 versus 114.324 GPU seconds (Table 4). Positive and negative count errors cancel in the total; their timing still matters to control. Together with the event-level counterexamples, this shows why neither startup correction nor target agreement alone establishes fidelity. The supplement retains all 36 prediction/physical pairs and 12 model/window means; this diagnosis neither changes M1 nor reselects a model. The later Bridge cells are a separate policy-contract study.

E's equal-window target disagreement remains 26.77% against the registered 25% expectation; M3 is diagnostic, not a veto. Predicted and observed switch counts match except one burst-offline repeat (22 predicted, 24 observed). These diagnostics define the next test: freeze the replay prediction, let a learned checkpoint drive the same physical lifecycle, and compare terminal feasibility and cost from the shared ledger.

Table 5. E2 policy execution on two A100 GPUs. P1+ requires every online and offline request to meet its deadline. Costs are registered-window scenario USD under the declared ledger. R/S report P1+ passes/runs in recovery/steady. The last column is cost reduction relative to all2; † marks a failed arm and is descriptive only.

Arm	R	S	Cost R	Cost S	Cost ↓ (%)
all2	1/1	1/1	4.200	4.200	0
C-1	3/3	3/3	2.357	2.449	42.8
C30-1	3/3	3/3	2.073	2.084	50.5
C30-2	3/3	3/3	2.051	2.103	50.5
C30-3	0/3	0/3	1.899	2.090	52.5 [†]
C-2	0/3	0/3	2.730	2.950	32.4 [†]

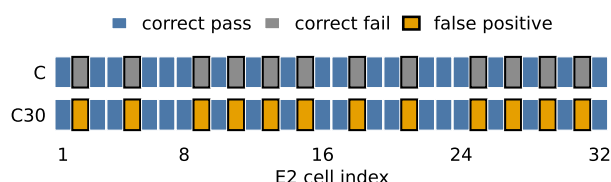


Figure 2. Replay-to-deployment feasibility for all 32 E2 cells. Blue and gray tiles are correct pass/fail directions; orange tiles are C30 false positives. Black outlines mark cells that physically fail P1+.

8.2 Policy execution and screening: feasibility is part of the result

E2 tests whether checkpoints selected in replay remain feasible in physical execution. In its re-provisioned runtime, 196 of 218 window restarts that reached readiness across the 30 PPO runs took over 60 s. Each window begins with an all2 capacity anchor; the five PPO arms then run in three repeats. All 32 cells are technically valid, and the package contains 36 attempts, including four preserved technical failures. C30-2 is the preregistered reference checkpoint. The table reports comparison means and one all2 run per window.

Three PPO arms pass all six physical repeats: C-1, C30-1, and C30-2; both all2 anchors also pass. The three feasible PPO arms have scenario costs 42.8–50.5% below all2, but this comparison is conditional on the declared ledger and synthetic API sink. C-2 is a decisive negative case: it averages only 4–7 on-time offline requests out of 120 and leaves about 80 unfinished, despite lower cost. C30-3 is a sharper boundary: it completes the offline stream eventually, but roughly 11 of 120 offline requests miss their deadline per repeat, so its lower cost is not feasible saving. Online requests are on time in every cell; the E2 failures are therefore isolated to the offline deadline path rather than hidden by an online regression.

Table 6. Frozen replay versus E2 physical outcomes. Agreement counts repeated physical cells; false positives predict feasibility where P1+ fails.

Replay	Agreement	False positives	Cost MAE
C	32/32	0/12	0.0804
C30	20/32	12/12	0.4657

Table 7. Replay-to-deployment agreement and window-cost MAE. Bridge and E2 include all technically valid cells, including anchors and deadline failures; E3 includes its 24 policy cells. Agreement compares each window prediction with repeated physical outcomes. Cost MAE is in scenario USD.

Campaign	Replay	Agreement	Cost MAE
Bridge	C	36/38	0.0969
Bridge	C30	21/38	0.6998
E2	C	32/32	0.0804
E2	C30	20/32	0.4657
E3	C	18/24	0.3112
E3	C30	24/24	0.0163

Predictions were frozen before physical execution. Table 6 compares each cell with its corresponding replay: C correctly predicts all 20 passes and 12 failures, whereas C30 predicts all cells feasible and misses both negative checkpoints. For each replay model, these 32 comparisons reuse 12 policy/window predictions; the physical repeats are not independent predictions. The two replay models have window-cost MAE of 0.0804 and 0.4657 scenario USD, respectively. C-2 exposes a startup-cooldown interaction: the 60 s cooldown begins at the capacity-target change. In C30 replay, readiness precedes cooldown expiry, leaving time to serve queued work; in C replay and most physical restarts, readiness follows expiry, allowing almost immediate scale-down. Across the six physical C-2 runs, 72 of 73 restarts taking over 60 s enter draining within two seconds of readiness (see the supplement’s Retrospective E2 lifecycle diagnosis section).

The earlier 38-cell Bridge campaign contains rule, guard, and PPO arms on a separate runtime. C also has higher feasibility agreement and lower cost MAE there (Table 7); the campaigns are not pooled. This repeated agreement motivates testing C as a screening rule on new checkpoints.

Both training environments yield passing and failing E2 checkpoints. E3 tests whether C’s agreement transfers to six previously undeployed checkpoints. From 90 eligible candidates, C predicts 67 feasible on both known windows and rejects 23. A fixed ordering selects three distinct trajectories from each group; C30 does not enter selection. All six pass every physical run (two windows, two repeats), as do both

Table 8. Formal held-out results. Each arm has three repeats per window; H has 15 offline deadline misses across its three recovery repeats.

Arm	tech.	offline	online
all2	9/9	9/9 (120/120)	9/9
all1	9/9	9/9 (120/120)	9/9
H	9/9	6/9	9/9
H+guard	9/9	9/9 (120/120)	9/9

all2 anchors. C therefore falsely rejects three feasible candidates, while C30 accepts all six. Across the 24 policy cells, C30 also has lower cost MAE (Table 7), reversing the E2 ordering. These selected cases show that earlier agreement does not establish reliable screening of new checkpoints and execution conditions; they do not estimate population screening accuracy. The supplement records candidate selection, predictions, and the separate window allocations.

8.3 Control case study: feasibility and cost

The earlier rule-based guard remains the control case for the same contract; E2 extends that contract to frozen learned checkpoints. Table 8 summarizes the rule-control case study. All2 and all1 satisfy the registered completion target. H is technically valid but fails the P1 gate in all three recovery repeats (116/120, 113/120, and 116/120 offline on time). In that held-out study, the guard repairs this failure without changing the workload or the locked model.

H+guard’s window cost is lower than all1 by 21.96%, 48.19%, and 58.08% in steady, recovery, and burst-offline, respectively. Their equal-window mean is 42.74%, above the preregistered 25% P2 line, although steady falls below that stretch target. Including setup and cleanup gives a separate mean saving of 38.86%.

The primary ledger covers the same 2,100 s interval; the deployment ledger also charges setup and cleanup. The supplementary table retains both views, with unchanged prices and event streams.

Before execution, E’s locked recovery prediction identifies four unfinished offline requests; the three physical H repeats leave four, seven, and four unfinished, for 15 total misses. This prediction is interpreted alongside, not substituted for, the observed completion evidence.

8.4 Latency–cost trade-off

The saving has a visible online latency price. The supplementary tail table reports each repeat’s online p99 rather than a pooled average. The always-on references remain near 1.6–2.6 s, while both dynamic arms reach about 15–19 s because requests can wait for the 10 s routing path and the synthetic API sink. The guard’s role is deadline protection

Table 9. Guard activity by recovery repeat. Ticks are window seconds; slack is the minimum pending-offline time to deadline (seconds), before subtracting service time.

Repeat	risk ticks	target raises	tick range	slack range
1	30	1	1982–2011	29–58
2	0	0	–	–
3	29	1	2027–2055	30–58

for offline work; it does not remove this routing delay. Every observed p99 remains below the registered 60 s online deadline, so the result is an explicit SLO-margin-for-cost trade-off.

8.5 Guard activation and mechanism

The guard checks deadline risk using the visible queue, running-job ages, lifecycle delay, and concurrency-four service estimate. Equation 3 retains risk IDs until dispatch or removal from the pending queue and distinguishes risk from a target raise. For the earlier guard case study, the formal logs contain 59 risk-active ticks, all in recovery: the guard raised the target above H twice, with the remaining risk ticks maintaining the already-raised target. No risk-active tick occurred in steady or burst-offline. Per-repeat counts and slack ranges are shown in Table 9.

8.6 Price sensitivity

We reprice H+guard and all1 over 27 window–GPU–API price combinations, keeping their event streams fixed. At the reference prices the three window savings are 21.96%, 48.19%, and 58.08%; the sign can reverse when GPU capacity is cheap and API use is expensive. The only negative cell is steady at GPU multiplier 0.5 and API multiplier 2.0, where savings are –39.59%. The steady break-even API multiplier at half-price GPUs is 1.004. Recovery and burst remain positive across the tested grid, but this is a finite sensitivity analysis, not a provider billing forecast.

External checks evaluate three implemented rule baselines: queue-target scaling (B-HPA), Holt queue prediction (B-PRED), and queueing-based capacity selection (B-MMC). Each meets all deadlines in its nine formal runs. A separate same-runtime comparison finds lower H+guard cost than B-HPA on two known Qwen windows, with higher API share and queue exposure. These results and the Llama and load-boundary studies bound the guard’s scope; they are not additional E2 samples. Full results and baseline definitions appear in the supplement’s External-validity evidence section.

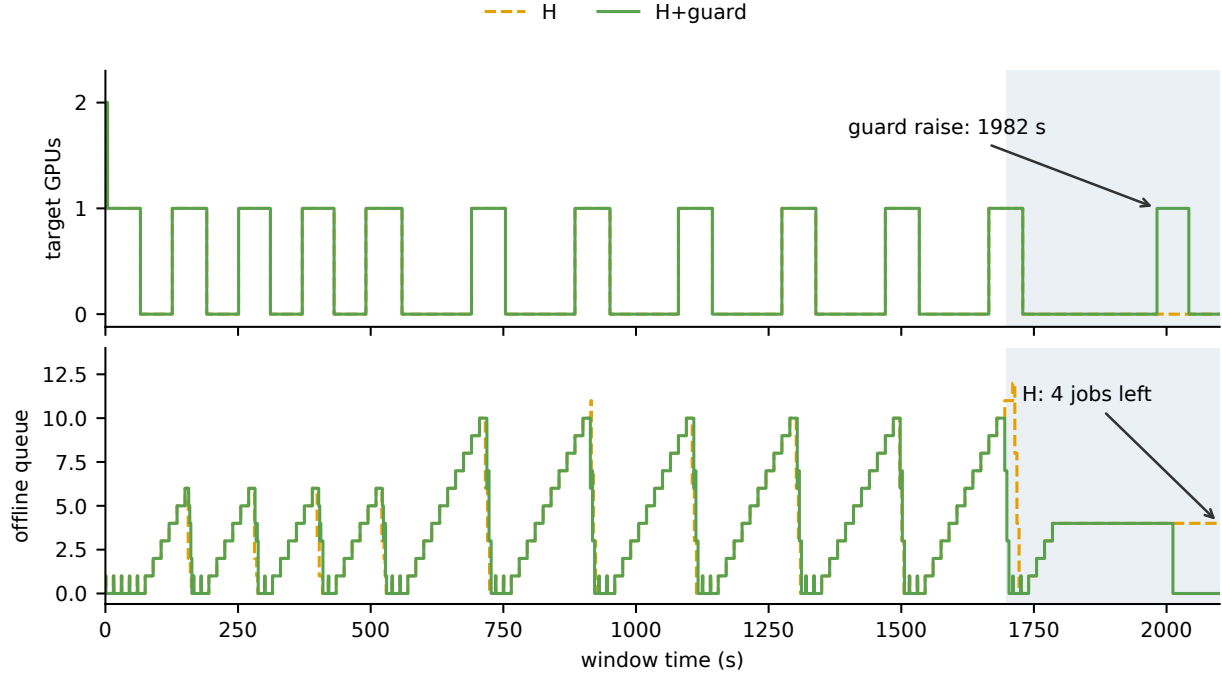


Figure 3. Recovery target and offline queue in repeat 1. The shaded final interval contains the guard raise at 1982 s; H leaves four offline jobs queued, while the guard clears them. Across all three H repeats, 15 jobs miss their deadlines.

Table 10. Price-grid savings (percent) for API multipliers 0.5/1.0/2.0. Rows are GPU multipliers 0.5/1.0/2.0; the final column is the break-even API multiplier.

Window	GPU	API .5	API 1	API 2	break-even
steady	0.5	20.05	0.17	-39.59	1.004
steady	1.0	31.90	21.96	2.07	2.104
steady	2.0	37.84	32.86	22.92	4.304
recovery	0.5	46.51	41.13	30.37	4.822
recovery	1.0	50.88	48.19	42.81	9.953
recovery	2.0	53.07	51.73	49.04	20.216
burst-offline	0.5	56.53	53.75	48.18	10.647
burst-offline	1.0	59.47	58.08	55.29	21.841
burst-offline	2.0	60.95	60.25	58.86	44.227

9 DISCUSSION

FidelityLoop evaluates whether policies meet deadlines and what resources feasible executions consume. C30 reduces fixed-H resource error, whereas C matches feasibility better in Bridge and E2; E3 then exposes C’s false rejections. An evaluator must therefore be checked on the deployment it is meant to screen.

9.1 Validating predictions across levels

Lifecycle validation should follow the controller’s clock. For controllers that block scale-down until a target-change cooldown expires, readiness after expiry may permit immediate scale-down even when replay predicts a service interval protected from scale-down. E2 exhibits this interaction. Comparing deployment startup timing with controller boundaries is therefore a useful diagnostic; setup and window scale-out should be measured separately. State and action trajectories also need inspection because aggregate resource error can hide temporal misalignment.

The guard shows how a controller can act on deadline risk from observed queues and frozen safety estimates. It repairs H’s held-out recovery failure, but H does not read the simulator’s startup parameters: the endpoint correction and the guard benefit have distinct causal roles.

9.2 Resource and latency trade-offs

The elastic policies trade online latency margin for lower occupied capacity: meeting the 60-second SLO can still leave excessive tail latency. The steady-window cost reversal at half-price GPUs and high API prices also shows that policy choice depends on the accounting assumptions.

The same-runtime baseline comparison makes this trade-off explicit. On two known Qwen inputs, H+guard lowers

cost relative to B-HPA while increasing synthetic-API share, lifecycle-event counts, and queue exposure. The comparison evaluates complete policies, including their dispatch rules. The Llama deadline miss and the failed higher-intensity Azure capacity tests show where the successful Qwen outcome stops transferring. On the first feasible tested Azure intensity, most savings occur in the final 600 seconds.

The supplementary CPU microbenchmark times `adapter.decide` on reconstructed states outside live serving. It measures decision-call latency, not full-controller or end-to-end overhead.

Practical validation. Before relying on predicted savings, check service timing, resource use, and control trajectories separately; freeze predictions before physical runs and condition savings on deadline feasibility, retaining failures; and recheck the evaluator in the intended deployment context. The paired traces make these checks inspectable on the recorded deployments, without assuming that earlier agreement will transfer.

10 LIMITATIONS AND NEGATIVE RESULTS

The synthetic API uses a five-second timer and a routing wait; it does not generate or evaluate an answer. Consequently, the cost result is a controlled resource contract, not a claim that a commercial API can replace local serving at equal quality. GPU off is verified process and resource release, not physical power-off, energy measurement, or a cloud provider bill. The three windows are cases rather than independent workloads, and the three repeats support engineering robustness, not broad statistical generalization.

H’s recovery misses show that economical cooldown decisions can leave offline work unfinished. The guard repairs the observed case without a general safety guarantee. The historical V-E comparison remains confounded by deployment drift; later refitting and policy comparisons do not isolate its calibration effect.

E2 covers two A100 devices, one Qwen model, two windows, and five frozen checkpoints. It establishes neither heterogeneous-device scalability nor a general ranking of PPO training environments or PPO versus the guard. The 38-cell Bridge campaign uses a separate runtime and is not pooled with E2. E3 adds six nonrandom checkpoints from two training seeds on known windows, with one allocation per window; checkpoint and execution-condition effects are not isolated. Its all-feasible cohort cannot measure detection of infeasible deployments or establish a generally superior replay model. Llama adds one similarly sized family and two repeats, including a guarded deadline failure; broader model scales, quantization, and engines remain untested. The same-runtime baseline compares two com-

plete policies on known Qwen inputs, so it neither adds held-out workloads nor isolates the guard inside the other controllers. Candidate families and workloads were locked before replay, with no window replacement. Missing session identities and regular offline pressure limit population-level inference. CPU verification covers all accepted records. A recovery addendum restores the root environment snapshot from identical, hash-bound per-run copies; the original archive checksum sidecar remains unavailable.

11 CONCLUSION

FidelityLoop provides a replayable contract for connecting elastic-serving policy decisions to lifecycle execution and resource/cost accounting. It shows why predictions need validation across levels: local-service fit can improve while resource prediction worsens, and accurate total occupation can hide large trajectory errors. Startup-endpoint isolation and event-level counterexamples explain where these checks diverge. Lifecycle states, control decisions, and resource use should therefore be validated together.

The guard case study quantifies deadline protection and its resource, routing, and cost trade-offs. E2 extends the same contract to frozen PPO checkpoints: some checkpoints reduce scenario cost while preserving every deadline, whereas the negative checkpoints reveal why cost must be conditioned on feasibility. E3 also exposes false rejections by the replay model that best matched E2. Under the studied synthetic-API setting, both the candidate policy and its screening model need deployment validation before predicted savings guide decisions.

REFERENCES

- Agrawal, A., Kedia, N., Mohan, J., Panwar, A., Kwatra, N., Gulavani, B. S., Ramjee, R., and Tumanov, A. Vidur: A Large-Scale Simulation Framework for LLM Inference. In *Proceedings of Machine Learning and Systems*, volume 6, pp. 351–366, 2024. URL https://proceedings.mlsys.org/paper_files/paper/2024/hash/b74a8de47d2b3c928360e0a011f48351-Abstract-Conference.html.
- Crankshaw, D., Sela, G.-E., Mo, X., Zumar, C., Stoica, I., Gonzalez, J., and Tumanov, A. InferLine: Latency-Aware Provisioning and Scaling for Prediction Serving Pipelines. In *Proceedings of the 11th ACM Symposium on Cloud Computing*, pp. 477–491, 2020. doi: 10.1145/3419111.3421285. URL <https://doi.org/10.1145/3419111.3421285>.
- Fu, Y., Xue, L., Huang, Y., Brabete, A.-O., Ustiugov, D., Patel, Y., and Mai, L. ServerlessLLM: Low-Latency Serverless Inference for Large Language Models. In *18th*

- USENIX Symposium on Operating Systems Design and Implementation (OSDI 24)*, pp. 135–153, 2024. URL <https://www.usenix.org/conference/osdi24/presentation/fu>.
- Gujarati, A., Karimi, R., Alzayat, S., Hao, W., Kaufmann, A., Vigfusson, Y., and Mace, J. Serving DNNs like Clockwork: Performance Predictability from the Bottom Up. In *14th USENIX Symposium on Operating Systems Design and Implementation (OSDI 20)*, pp. 443–462, 2020. URL <https://www.usenix.org/conference/osdi20/presentation/gujarati>.
- Kabakibo, H. S., Trivedi, A., and Wang, L. Breaking the Ice: Analyzing Cold Start Latency in vLLM. In *Proceedings of Machine Learning and Systems*, volume 8, pp. 1187–1205, 2026. URL https://proceedings.mlsys.org/paper_files/paper/2026/hash/29416b66c2149872b9d1415a3fd2c5e0-Abstract-Conference.html.
- Kwon, W., Li, Z., Zhuang, S., Sheng, Y., Zheng, L., Yu, C. H., Gonzalez, J., Zhang, H., and Stoica, I. Efficient Memory Management for Large Language Model Serving with PagedAttention. In *Proceedings of the 29th Symposium on Operating Systems Principles*, pp. 611–626, 2023. doi: 10.1145/3600006.3613165. URL <https://doi.org/10.1145/3600006.3613165>.
- Patel, P., Choukse, E., Zhang, C., Shah, A., Goiri, Í., Maleki, S., and Bianchini, R. Splitwise: Efficient Generative LLM Inference Using Phase Splitting. In *2024 ACM/IEEE 51st Annual International Symposium on Computer Architecture (ISCA)*, pp. 118–132, 2024. doi: 10.1109/isca59077.2024.00019. URL <https://doi.org/10.1109/isca59077.2024.00019>.
- Schulman, J., Wolski, F., Dhariwal, P., Radford, A., and Klimov, O. Proximal Policy Optimization Algorithms, 2017. URL <https://arxiv.org/abs/1707.06347>.
- Sun, B., Huang, Z., Zhao, H., Xiao, W., Zhang, X., Li, Y., and Lin, W. Llumnix: Dynamic Scheduling for Large Language Model Serving. In *18th USENIX Symposium on Operating Systems Design and Implementation (OSDI 24)*, pp. 173–191, 2024. URL <https://www.usenix.org/conference/osdi24/presentation/sun-biao>.
- Wang, Y., Chen, Y., Li, Z., Kang, X., Tang, Z., He, X., Guo, R., Wang, X., Wang, Q., Zhou, A. C., and Chu, X. BurstGPT: A Real-world Workload Dataset to Optimize LLM Serving Systems, 2024. URL <https://arxiv.org/abs/2401.17644v3>.
- Yang, M., Zhang, Z., Wu, M., Yan, J., Sun, H., and Chang, L.-W. Charon: A Unified and Fine-Grained Simulator for Large-Scale LLM Training and Inference. In *Proceedings of Machine Learning and Systems*, volume 8, pp. 94–107, 2026. URL https://proceedings.mlsys.org/paper_files/paper/2026/hash/dbc8ce0fdcd55172d73fb05dbae07fc-Abstract-Conference.html.
- Zhang, C., Yu, M., Wang, W., and Yan, F. MArk: Exploiting Cloud Services for Cost-Effective, SLO-Aware Machine Learning Inference Serving. In *2019 USENIX Annual Technical Conference*, pp. 1049–1062, 2019. URL <https://www.usenix.org/conference/atc19/presentation/zhang-chengliang>.
- Zhang, G., Guo, W., Tan, Z., Guan, Q., and Jiang, H. KISS: A GPU-Aware Kubernetes Inference Simulator with RL-Based Auto-Scaling, 2025a. URL <https://arxiv.org/abs/2507.07932>.
- Zhang, G., Vippagunta, S., Nandagopal, R., Raman, S., Xu, J., Pfeiffer, M., Chatterjee, S., Tan, Z., Guo, W., and Jiang, H. AAPA: An Archetype-Aware Predictive Autoscaler with Uncertainty Quantification for Serverless Workloads on Kubernetes, 2025b. URL <https://arxiv.org/abs/2507.05653>.
- Zhang, H., Tang, Y., Khandelwal, A., and Stoica, I. SHEPHERD: Serving DNNs in the Wild. In *Proceedings of the 20th USENIX Symposium on Networked Systems Design and Implementation*, pp. 787–808, 2023. URL <https://www.usenix.org/conference/nsdi23/presentation/zhang-hong>.